

Real-Time Augmented Reality Visual-Captions for Deaf and Hard-of-Hearing Children in Classrooms

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ABSTRACT

Deaf and hard-of-hearing (DHH) children experience difficulties in mainstream classrooms since they cannot access audio information effectively. Although tools like captions are developed specifically to assist DHH individuals, primary school children have trouble to read or understand the text. To address this challenge, this paper develops an AR-based visual-captions, which will transfer the speech to text and images in real-time and display the information around the teacher in classrooms. The AR visual-captions aims to help DHH children better receive information and enhance their classroom experience. We describe the concept, design and implementation of our prototype, and discuss future research directions.

Keywords: Augmented Reality, Accessibility, DHH, Children, Captions.

Index Terms: K.6.1 [Management of Computing and Information Systems]: Project and People Management—Life Cycle; K.7.m [The Computing Profession]: Miscellaneous—Ethics

1 INTRODUCTION AND RELATED WORK

Over 5% of the world's population (430 million people) have hearing loss, among them, 34 million are children [10]. It is estimated that the number of deaf or hard of hearing (DHH) individuals will reach 700 million by 2050 [10]. DHH individuals are facing challenges that limit their ability to participate in areas such as education [1]. One of the educational challenges is the difficulty for DHH people to access audio information in the mainstream classroom in the same way as hearing individuals [6,7]. DHH people usually benefit from assistive devices such as hearing aids [10]. However, current production of hearing aids could not always meet the needs [1]. Captions are the most widely used approaches to show the recognized text [3,4,9,10]. However, these methods may pose additional limitations. The captions that are shown on a separate screen might split the user's visual attention between the screen and the speaker [3,7].

With the increasing popularity and accessibility of Augmented Reality (AR), accessibility research has focused on using AR to enhance the communication experience for DHH individuals [9]. AR is changing the way of how information is presented to users by displaying virtual information in the real world without compromising users' natural vision [8]. With AR, the captions can be seamlessly integrated within the real world, displaying the captions directly in the user's field of view could reduce the visual attention split and will not obstruct users to see the real environment [3,4]. SpeechBubbles [9] is an AR interface to improve DHH individuals' experience of comprehending speech

from hearing individuals in face-to-face group conversations. AR Windows [3] displays real-time AR captions for DDH individuals, allowing users to manipulate the shape and placement of captions in 3D space.

However, AR captions are still challenging for DHH children in primary school since they might have limited comprehension skills for textual information and the captioning speed is likely to exceed their reading speed [6,7]. Sign language interpreters might be a solution, and Luo et al. [6] have developed an AR interface for synchronized sign language interpreters for DHH students in classrooms, but it heavily relies on actor performances and advance recordings. Liao et al. [5] have presented a novel approach to display keywords and visual elements to facilitate interactive storytelling experience, while the system mainly focuses on the perspective of presenters creating augmented live presentations. Although AR has been reported with advantages in special education to facilitate learning for students with special needs, little has been done for DHH children, especially in designing captioning systems in mainstream classrooms.

In this study, we designed an AR-based interface to transfer the speech of the teacher to text and extract the keywords to visuals in real-time to facilitate DHH children in primary school to learn and understand more effectively in mainstream classrooms. We describe the concept, design choices and implementation of our prototype, and how the prototype can be applied to help DHH children. Finally, we present research directions for future study.

2 DESIGN AND IMPLEMENTATION

2.1 Concept

Our idea is to provide real-time visual-captions and take the advantages of AR technology to display the virtual information in the real world environment for DHH children to help them better understand speech information in the classroom. The AR interface first provides real-time speech-to-text captions that can be clearly presented to children. Second, the AR interface presents visuals searched from Google Image using the keywords that are automatically extracted from the recognized speech text. Third, the captions and the visuals will be continuously displayed around the teacher via AR devices. The virtual content is located in spatial position and will not block the user's view. For example, in the classroom setting, when the teacher is moving around the classroom, the captions and the visuals will follow his/her head/body.

2.2 Design Choice

Location of visual-captions: Previous study has shown that for AR visual displays, the text position is an important factor with respect to the readability [2]. Users would prefer that captions could automatically align close to the speaker to reduce the visual attention split [4]. In the meantime, when the speaker is moving out of the user's field of view, the user should still receive the information as on-screen captions. Therefore, in our current prototype, the visual-captions will always follow the teacher and be displayed in spatial position. If the speaker walks out of the user's field of view, the captions and the visuals will be

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automatically transferred to on-screen subtitles and images, and will be changed back when the teacher is back to the user's field of view.

Appearance of visual-captions: In prior work, there is no unified design of the colors, size, fonts, etc. of the captions. In AR settings, the environment in the real world is changing in different situations. That might be the reason why the appearances are different in different studies. Some use white background with black text [e.g., 9] and some use white text without background [e.g., 3]. In our prototype, we use a white background and black text by default. See Figure 1 top. In the meantime, the user is also provided with different choices of colors and fonts of the text and can customize by themselves. See Figure 1 bottom. The size of the captions and visuals can be adjusted by their own needs via the AR device.

Device: Most prior works have used HoloLens as the AR device while issue has been identified and a lighter and smaller device is needed [3]. HoloLens is also too heavy for primary school children. Therefore, in the current study, we use mobile devices (iPad or iPhone) to run the prototype. The mobile device is put on a holder on top of the user's desk, facing to the front of the classroom (teacher).

2.3 Implementation

Our AR prototype leverages the following components:

Speech recognition: We recognize and transcribe the speech of the teacher into text in real-time with Vosk toolkit in Unity, which provides the ability to recognition voice context from the microphone input on the device for more than 20 languages. The text transcriptions will be displayed as AR captions and used for word extraction in following steps. The current language used in the prototype is English.

Word extraction: We use Python script on SpaCy, a natural language processing model, to detect and extract the words from the speech text, using the pre-trained model based on the model of *en_core_web_md_dataset*. Only nouns will be extracted, which will then be used for visual searching next.

Visual search: The visuals are collected using GoogleImageCrawler with the nouns extracted in the last step. The first result on Google Image will be obtained and can be used for displaying in Unity later.

Visual display: We build the AR interface in Unity to display the captions and visual elements based on ARKit and AR Foundation around the speaker. The captions will be displayed on



Figure 1: AR visual-captions with black text and with 3D text (images obtained from Google Image with the keyword *playground* and *slide*).

top of the speaker's head with the AR face tracking or body tracking and the visual elements will be displayed on the left or right side around the captions.

3 CONCLUSION AND FUTURE WORK

In this paper, we present AR-based visual-captions, a real-time speech to textual/visual elements to help DHH children improve their classroom experience. We describe the concept, the design rationales, and the implementation of the prototype on mobile devices. The prototype still has limitations such as the randomness of the obtained images from Google. In the future, we will refine the design and conduct further user studies to evaluate the usability of the prototype. We also plan to extend the current concept to provide feedback to teachers and allow group conversations. For example, we can collect the eye tracking data of children following the captions and visuals, and give the feedback to the teacher so that they can pay more attention to their speaking speed or their locations in the classroom.

REFERENCES

- [1] A. Alnafjan, A. Aljumaah, H. Alaskar, and R. Alshraihi. Designing "Najeeb": Technology-enhanced learning for children with impaired hearing using Arabic sign-language ArSL Applications. In 2017 International Conference on Computer and Applications (ICCA) (pp. 238-273). IEEE. 2017.
- [2] S. Eksvård and J. Falk. Evaluating Speech-to-Text Systems and AR-glasses: A study to develop a potential assistive device for people with hearing impairments. 2021.
- [3] D. Jain, B. Chinh, L. Findlater, R. Kushalnagar, and J. Froehlich. Exploring augmented reality approaches to real-time captioning: A preliminary autoethnographic study. In Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems (pp. 7-11). 2018.
- [4] D. Jain, R. Franz, L. Findlater, J. Cannon, R. Kushalnagar, and J. Froehlich. Towards Accessible Conversations in a Mobile Context for People who are Deaf and Hard of Hearing. In Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility (pp. 81-92). 2018.
- [5] J. Liao, A. Karim, S.S. Jadon, R.H. Kazi and R. Suzuki. RealityTalk: Real-Time Speech-Driven Augmented Presentation for AR Live Storytelling. In Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (pp. 1-12). 2022.
- [6] L. Luo, W. Dongdong, S. Guo, H. Jie, and T. Ziqi. Avatar Interpreter: Improving Classroom Experiences for Deaf and Hard-of-Hearing People Based on Augmented Reality. In CHI Conference on Human Factors in Computing Systems Extended Abstracts, pp. 1-5. 2022.
- [7] M. Marschark, J.B. Pelz, C. Convertino, P. Sapere, M.E. Arndt, and R. Seewagen. Classroom interpreting and visual information processing in mainstream education for deaf students: Live or Memorex@?. American Educational Research Journal, 42(4), pp.727-761. 2005.
- [8] D.X. Ong, K.X., Chia, Y.Y. Huang, J.T.S. Teo, J. Tan, M. Lim, D. Qiu, X. Xia, and F.Y. Guan. Smart Captions: A Novel Solution for Closed Captioning in Theatre Settings with AR Glasses. In 2021 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI) (pp. 1-5). IEEE. 2021.
- [9] Y.H. Peng, M.W. Hsi, P. Taele, T.Y. Lin, P.E. Lai, L. Hsu, T.C. Chen, T.Y. Wu, Y.A. Chen, H.H. Tang, and M.Y. Chen. Speechbubbles: Enhancing captioning experiences for deaf and hard-of-hearing people in group conversations. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (pp. 1-10). 2018.
- [10] World Health Organization. Deafness and hearing loss. Available: <https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss>. 2021.